

GEOLOGICAL SURVEY & MINERAL RESERVE REPORT

Kharpatia Manganese Ore Block, Sundargarh District, Odisha

Prepared for the Directorate of Mines, in accordance with Rule 12 of the Mineral Conservation and Development Rules

Report No. GSR/OR/1987-88/114

Survey Season: Nov 1987 - Feb 1988

Prepared by: Geological Survey Party No. 6, Regional Office, Bhubaneswar

1. INTRODUCTION

This report presents the findings of a detailed geological mapping and pitting programme carried out over the Kharpatia block during the 1987-88 field season. The block covers an area of approximately 4.6 sq. km within the Bonai-Keonjhar manganese belt and is underlain by banded ferruginous formation of the Iron Ore Group, with manganese ore occurring as lenticular bodies along strike-parallel shear zones. A total of eleven pits and four bore holes were sunk to establish the continuity and grade of the principal ore body, hereinafter referred to as the 'Main Lode'.

2. RESERVE ESTIMATE

Ore reserves have been classified as Proved, Probable and Possible following the criteria laid down by the Indian Bureau of Mines. Tonnage figures are computed on a specific gravity of 3.7 for manganese ore, using the cross-sectional method between adjacent pit lines. The summary below sets out reserve tonnage and average grade by category.

Table 1: Ore Reserve Summary by Category

Category	Tonnage (MT)	Avg. Mn %	Avg. Fe %	SiO ₂ %
Proved	212,480	38.6	9.2	11.4
Probable	164,900	36.1	10.7	13.8
Possible	97,350	33.4	12.1	16.2
Total	474,730	36.5	10.5	13.5

3. DRILL-HOLE ASSAY SUMMARY

Core recovery averaged 82 percent across the four diamond bore holes (DDH-1 to DDH-4). Assay intervals were composited over true ore width, weighted by recovered core length; results are given in Table 2 overleaf, alongside monthly ore-raising figures for the adjoining working pits in Table 3.

Table 2: Diamond Drill-Hole Assay Results

Hole No.	From (m)	To (m)	Width (m)	Fe %
DDH-1	14.20	22.80	8.60	8.9
DDH-2	9.40	15.10	5.70	10.4
DDH-3	18.00	27.35	9.35	7.8
DDH-4	11.60	16.90	5.30	11.6

Table 3: Monthly Ore Raised - Working Pits (1987)

Month	Ore Raised (t)	Mn Grade %	Waste (t)	Remarks
August	3,140	37.8	5,860	Normal
September	2,975	38.1	5,410	Normal
October	3,402	36.4	6,120	Monsoon tail
November	3,610	37.0	6,340	Normal

4. REMARKS AND RECOMMENDATIONS

The Main Lode remains open along strike to the north-east and is recommended for further extension drilling in the next field season. Grade continuity between DDH-2 and DDH-4 should be treated with caution owing to poor core recovery in the intervening interval, and confirmatory pitting is advised before these blocks are upgraded to the Proved category.

Submitted by,

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